

# 1 Configuration of Inspection Planning

## 1.1 Inspection plan header

The inspection plan header includes four different options significantly affecting the generation of inspection plans and inspection requirements.

| Operation                     |                                                    |                                                                           |
|-------------------------------|----------------------------------------------------|---------------------------------------------------------------------------|
| Operation assignment          | <input type="radio"/> one insp. plan for each OP   | <input checked="" type="radio"/> one inspection plan for all OP           |
| IO + insp. station            | <input type="radio"/> an IO for all insp. stations | <input checked="" type="radio"/> an IO for each insp. station             |
| Generate new QM OPs           | <input checked="" type="radio"/> none              | <input type="radio"/> During creating new insp. requirement               |
| IO + generate characteristics | <input type="radio"/> During operation log on      | <input checked="" type="radio"/> During creation of new insp. Requirement |

### 1.1.1 Operation assignment

You can choose from the two assignment types "One inspection plan for each operation" and "One inspection plan for all operations". By assigning operations, you can specify whether the characteristics of various operations ("One inspection plan for all operations") shall be included in the inspection plan to be generated, or whether the inspection plan shall only include the characteristics of one operation.

#### "One inspection plan for all operations"

By default, you can only use the setting "One inspection plan for all operations". With this setting, you have to enter all characteristics to be inspected in one or more operations of this article/group in the inspection plan.

You can assign a characteristic to an operation directly in the inspection plan characteristic:

| Characteristic             | Properties                                                                                                        | Details | Specifications | Chart 1 | Default values chart 1 | Chart 2 | De |
|----------------------------|-------------------------------------------------------------------------------------------------------------------|---------|----------------|---------|------------------------|---------|----|
| Characteristic             |                                                                                                                   |         |                |         |                        |         |    |
| Seq.                       | 10                                                                                                                |         |                |         |                        |         |    |
| Characteristic             | 0001 ...                                                                                                          |         |                |         |                        |         |    |
| Characteristic designation | total length                                                                                                      |         |                |         |                        |         |    |
| Machine                    | ...                                                                                                               |         |                |         |                        |         |    |
| Group                      | ...                                                                                                               |         |                |         |                        |         |    |
| Planned on                 | <input checked="" type="radio"/> Machine <input type="radio"/> Group                                              |         |                |         |                        |         |    |
| Characteristic type        | <input type="radio"/> attributive <input type="radio"/> inspection hint <input checked="" type="radio"/> variable |         |                |         |                        |         |    |
| Inspection result base     | <input type="radio"/> all samples <input type="radio"/> last sample                                               |         |                |         |                        |         |    |
|                            | <input type="checkbox"/> Mandatory inspection                                                                     |         |                |         |                        |         |    |
|                            | <input type="checkbox"/> Formula                                                                                  |         |                |         |                        |         |    |
| Inspection station         | 07 ...                                                                                                            |         |                |         |                        |         |    |
| Operation                  | 0100 ...                                                                                                          |         |                |         |                        |         |    |
| Operation designation      |                                                                                                                   |         |                |         |                        |         |    |
| Initial sample creation    | <input type="checkbox"/> Copy characteristic                                                                      |         |                |         |                        |         |    |

### "One inspection plan for each operation"



In order to be able to use the setting "One inspection plan for each operation", the system must be customized by MPDV.

This setting results in the generation of a separate inspection plan for each operation of an item whose characteristics are to be inspected. The dialog now shows the fields "Operation" and "Operation designation". You can enter the operation number directly in the inspection plan header.

| Operation                     |                                                                                                                         |
|-------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| Insp. plan + OP               | <input checked="" type="radio"/> one insp. plan for each OP <input type="radio"/> one inspection plan for all OP        |
| IO + insp. station            | <input type="radio"/> an IO for all insp. stations <input checked="" type="radio"/> an IO for each insp. station        |
| Operation                     | ...                                                                                                                     |
| Operation des.                |                                                                                                                         |
| Generate new QM OPs           | <input checked="" type="radio"/> none <input type="radio"/> During creating new insp. requirement                       |
| IO + generate characteristics | <input type="radio"/> During operation log on <input checked="" type="radio"/> During creation of new insp. Requirement |



You may only indicate the operation designation if solely HYDRA-CAQ is licensed. In all other cases, you may only store the operation number. (This field must be used if operations are

processed!)

#### Please note



If possible, you should use the setting "One inspection plan for all operations". This, for instance, allows for printing all inspection characteristics of an item even if it comprises several operations.



In areas without any allocation to a production order, you should use the option "One inspection plan for all operations". The advantage here is that the "fictional" operation must be entered in the inspection plan header once only, instead of having to be entered for each inspection plan characteristic. The system must be configured accordingly by MPDV Consulting in order to use this inspection plan option.

#### Operation, operation designation

If operations are used, you only have to complete one of the two fields "Operation" or "Operation Designation". If an inspection requirement/inspection order shall automatically be created by the logon of an operation belonging to a production order, you only have to enter the operation number in the field "Operation". Please note that these fields are used as search criteria in the subsequent generation of an inspection order. This means if only the designation is entered for the operation in the inspection plan, you must also enter this precise information when creating the inspection order.



Even in areas without any allocation to a production order (e.g. in goods receipt and test equipment management), you must assign a "fictional" operation (e.g. 9999). This is necessary if inspection steps are generated.

#### 1.1.2 IO (inspection order) + inspection station

This option defines the procedure of generating inspection steps. You can choose whether "One inspection step (=IO) for each inspection station" or "One inspection step (=IO) for all inspection stations" is created.

##### Basis: Assignment of an inspection station to a characteristic

By assigning an inspection station to a characteristic, you can generate inspection steps which comprise only the characteristics of one inspection station, each. The decisive criterion for this is the configuration setting in the inspection plan header.

#### "One inspection step (=IO) for each inspection station"

If one inspection step is created for each inspection station, this information can in turn be used as a filter for the inspection orders available at the AIP inspection terminal. In addition, the terminal has to be configured accordingly.

### "One inspection step (=IO) for all inspection stations"

If you specified in the inspection plan header that an inspection step is not to be generated for each inspection station, the inspection station assigned to a characteristic only represents supplementary information.

#### 1.1.3 Generate new QM OPs

If you use this option, you can specify whether or not additional QM operations are produced during the generation of an inspection requirement.

You should choose "none" for in-production inspections, since the operations of a production order already exist.

You should select the setting "when generating new inspection requirements" for areas with no assignment to a production order; e.g. goods receipt, test equipment management. In these areas, inspection requirements are triggered by other processes (e.g. ERP interface, order generation in test equipment management, manual generation).

This setting generates separate QM operations when inspection requirements are generated. These OPs are to be logged in to the terminal in order to perform the inspection.



In areas with no assignment to a production order (e.g. in goods receipt and in test equipment management), a "fictional" operation is to be assigned (e.g. 9999). This is necessary if you want to generate inspection steps later on.

#### 1.1.4 IO + generate characteristics

This option specifies the time when an inspection step with the associated characteristics is generated:

- **With inspection requirement generation:** The appropriate inspection steps are generated at the same time as the inspection requirement.
- **With operation logon:** The inspection steps are only generated when an operation is logged on to the terminal. Only then are the specifications for the individual characteristics identified.

You can select the option "With operation logon", for example, if an operation is not regularly included in a production order. The inspection step for this operation is only generated if the OP is logged on. As a consequence, you do not have to complete the inspection step manually if this operation is not included in the production order.



In addition, this setting makes sense if, for example, an extended period has expired between the first logon of the order (typically logon of the first operation (= time of inspection requirement generation) and the first logon of a subsequent operation. Specifications might have changed and the modified ones are to be used for the subsequent operation.

### 1.1.5 Recommended basic settings for each area

- In this regard, you should consider various combinations of the 4 options, since some of them have an impact on each other and some combinations are not appropriate.  
In addition, you should take into account the recommended "basic settings" for each field of application (production, goods receipt, goods issue, test equipment calibration, initial sample, etc.). For instance, you should enable the setting for the generation of QM operations for goods receipt.

#### In-production inspection

- Operation assignment: [one inspection plan for all operations](#)
- IO + inspection station: [one IO for each inspection station \(if inspection stations are used\)](#)
- Generate new QM OPs: [none](#)
- IO + generate characteristics: [when generating inspection requirement](#)
- IO + generate characteristics: [when generating inspection requirement](#)

#### Initial sample inspection

- Operation assignment: [one inspection plan for all operations](#)
- IO + inspection station: [one IO for each inspection station \(if inspection stations are used\)](#)
- Generate new QM OPs: [when generating new inspection requirements](#)
- IO + generate characteristics: [when generating inspection requirement](#)

#### Goods receipt inspection

- Operation assignment: [one inspection plan for all operations](#)
- IO + inspection station: [one IO for all inspection stations \(normally only one inspection station is used in goods receipt\)](#)

- Generate new QM OPs: [when generating new inspection requirements](#)
- IO + generate characteristics: [when generating inspection requirement](#)

### Goods issue inspection

- Operation assignment: [one inspection plan for all operations](#)
- IO + inspection station: [one IO for all inspection stations \(normally only one inspection station is used in goods issue\)](#)
- Generate new QM OPs: [when generating new inspection requirements](#)
- IO + generate characteristics: [when generating inspection requirement](#)

### Test equipment management

- Operation assignment: [one inspection plan for all operations](#)
- IO + inspection station: [one IO for all inspection stations \(normally only one inspection station is used in test equipment management\)](#)
- Generate new QM OPs: [none](#)
- IO + generate characteristics: [when generating inspection requirement](#)

## 1.1.6 Properties

The screenshot shows the 'Properties' tab of the SAP Inspection Planning configuration. The 'Inspection type' is set to 'charact. related inspection'. The 'Action' is set to 'creation of insp. order'. The 'Gage method' is empty. The 'Dynamic modification' is set to 'None'. The 'Cavity assignment' is set to 'none'.

The tab of general inspection plan details includes further inspection plan characteristics:

- Inspection type (characteristic or piece-related) and
- Action (create and immediately release inspection step, or only create inspection step and release it manually at a later point in time).

The inspection plan characteristic "Inspection type" defines the subsequent inspection sequence.

Option 1: record the measured values of the first characteristic for all items to be inspected, and then change the characteristic and again record the measured values for all items (characteristic-related inspection - the item changes with every recorded measured value).

Option 2: Check each item separately and completely (piece-related inspection - the characteristic is changed after each recorded measured value).

The inspection plan characteristic "Action" defines whether the inspection plans generated based on this inspection plan shall be released immediately or manually. The option "Generate inspection step" only makes sense if specific information yet to be assigned to the inspection step is missing at the time when the inspection step is being generated.

The gage method is only available for the gage management.

You only have to specify dynamic modification for goods receipt and goods issue. You can define whether dynamic modification takes place in relation to a characteristic or in relation to a batch.

Normally the cavity assignment is only used in the production inspection.

## 1.2 Inspection plan characteristics

### Machine / Group

|            |                                                                      |
|------------|----------------------------------------------------------------------|
| Machine    | <input type="text"/>                                                 |
| Group      | <input type="text"/>                                                 |
| Planned on | <input checked="" type="radio"/> Machine <input type="radio"/> Group |

During inspection planning, you must enter a machine (a workplace) or a group for goods receipt, goods issue, initial sampling and calibration (test equipment), because true QM operations are generated in these areas.

If true QM operations are also to be generated in production, you have to assign a machine or a group to these characteristics. The assignment is necessary for true QM operations, because the sequencing list for logging in operations to the terminal only shows operations planned at the corresponding machine or group.



If CTWIN is used instead of AIP, machine or group assignment is omitted.



If there are identical operations, a separate inspection step is generated for each combination "Machine" or "Machine group" with a single "Inspection station".

### Possibilities of planning QM operations by graphic planning/order sequencing

By logging on the first operation of an order, the system generates the inspection requirement and all inspection steps, depending on the inspection plan configuration. As a consequence, the QM operations are generated, too, if this is defined in the inspection plan.

These QM operations may then be planned from the group to individual workplaces and/or other workplaces in the graphic planning/order sequencing.

The generation of the inspection requirement and hence the QM operations directly after transmission of the production order to HYDRA or in case of a specific status change of an operation (e.g. from not ready to prepared) is also possible. These individual logics have to be specified for each individual customer in the project.

### Inspection result base

The settings for the "inspection result base" define whether all or only the last sample is used for the characteristic result.

Inspection result base ☐ all samples ☒ last sample

#### If inspection points are used

- All samples: all samples, i.e. all inspection points, are used for the characteristic result.
- Last sample: only the last inspection point where inspection data was recorded for the characteristic is used for the characteristic result.



Special case: in a cavity-related inspection, several samples (= several inspection points) exist for a characteristic-wise inspection. For this reason, the setting 'Last sample' is not recommended and not correct for these characteristics.

#### If no inspection points are used

- All samples: all samples are used for the characteristic result.
- Last sample: only the last sample is used for the characteristic result.

## Mandatory inspection

☐ Mandatory inspection

### If inspection points are used

- Variable characteristics: the first measured value of the characteristic is shown in yellow for each inspection point. The other measured values are shown in light green.
- Attributive / inspection chart: the characteristic is shown in yellow for each inspection point.

### If no inspection points are used

- The first value to be recorded / the first sample of the characteristic is shown in yellow, regardless of the sample number.

## Calculated characteristics

|                                             |
|---------------------------------------------|
| <input checked="" type="checkbox"/> Formula |
| Formula <input type="text"/>                |

No context (e.g. machine, cavity, etc.) is transmitted to calculated characteristics. The reason is that the values to be used in the calculation are, for example, recorded on different machines. This results in an ambiguity as to which value of the calculated characteristic is to be assigned to which machine.